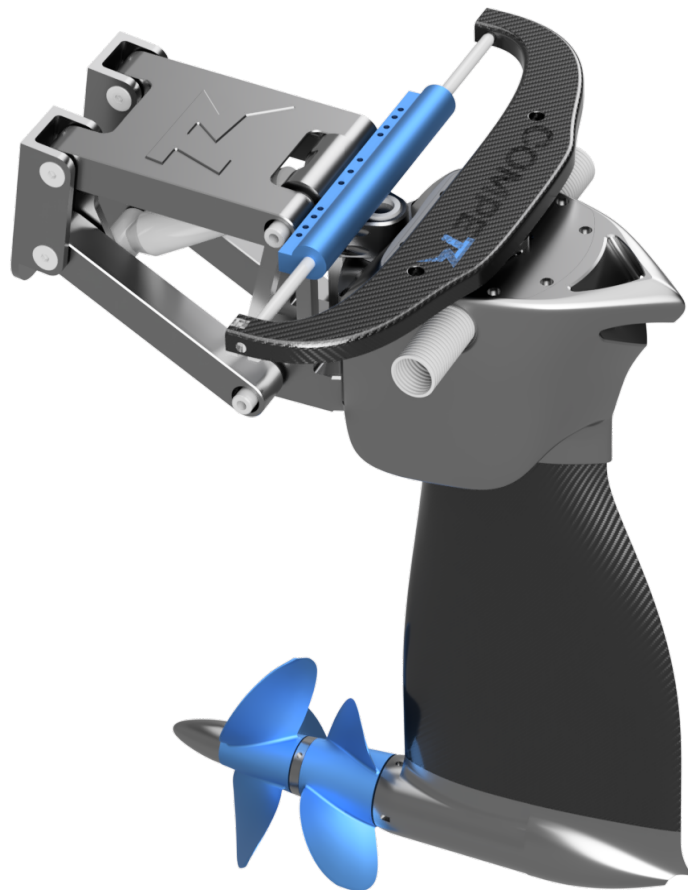


COMPET^{TR}

TECHNICAL DATASHEET



Summary

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1. General Specifications

Item	Value	Notes
Outboard		
Nominal Power	26.9kW (40HP)	
Maximum Power	42kW	
Maximum Torque	100Nm	
Propeller	Contra rotating	
Shaft length	500mm	Customizable, subject to MOQ
Motor	Axial flux PMSM, water cooled	
Dimensions	300x210x800mm	
Weight	25kg	
Battery		
Nominal voltage	96V	
Maximum voltage	109V	100% SoC
Maximum current	375A	Continuous
Nominal capacity	26kWh	
Configuration	26S	
BMS	Integrated	With continuous isolation monitoring
Communication	2x CAN-BUS 2.0B	CAN1: VCU and Charger CAN2: PC communication
Cooling	Passive	
Aux DC/DC	12V 1kW	Available to power on-board low voltage systems.
Dimensions	960x330x430mm	
Weight	192kg	
Protection grade	IP65	
Trim/Steering		
Trim type	Hydraulic, electroactuated	
Trim angle	-5°/+30°	
Trim assembly weight	10kg	
Steering type	Hydraulic, electroactuated	
Steering angle	-40°/+40°	

2. Electronics Specifications

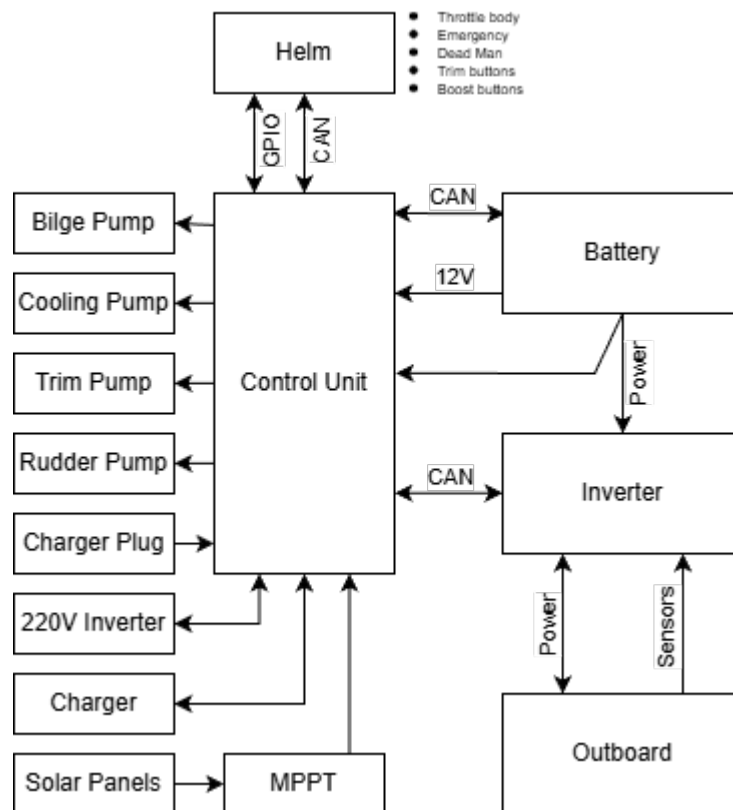
Item	Value	Notes
Control unit		
Supply voltage	12V	
Trim pump driver	H-Bridge	12V 40A max
Rudded pump driver	H-Bridge	12V 40A max
Bilge pump driver	Low side	12V 15A max. Bilge pump sensor through GPIO
Cooling pump driver	Low side	12V 15A max.
Charging	220V 3kW*	IEC 62196 Type 2 compliant.
Outputs	3x low side	12V 1A max
GPIO	3x analog/digital GPIO	
Communications	2x CAN 2.0B	
Connectors	1x 35-way sealed connector 1x 23-way sealed connector	
Dimensions	260x160x90	IP67
Weight	TBD	

Control unit board may be extended on request by adding GPS/GSM/BLE/WiFi connectivity for remote monitoring and fleet management. Contact Competr for further details.

Specifications subject to change without notify.

* Faster charge available on request

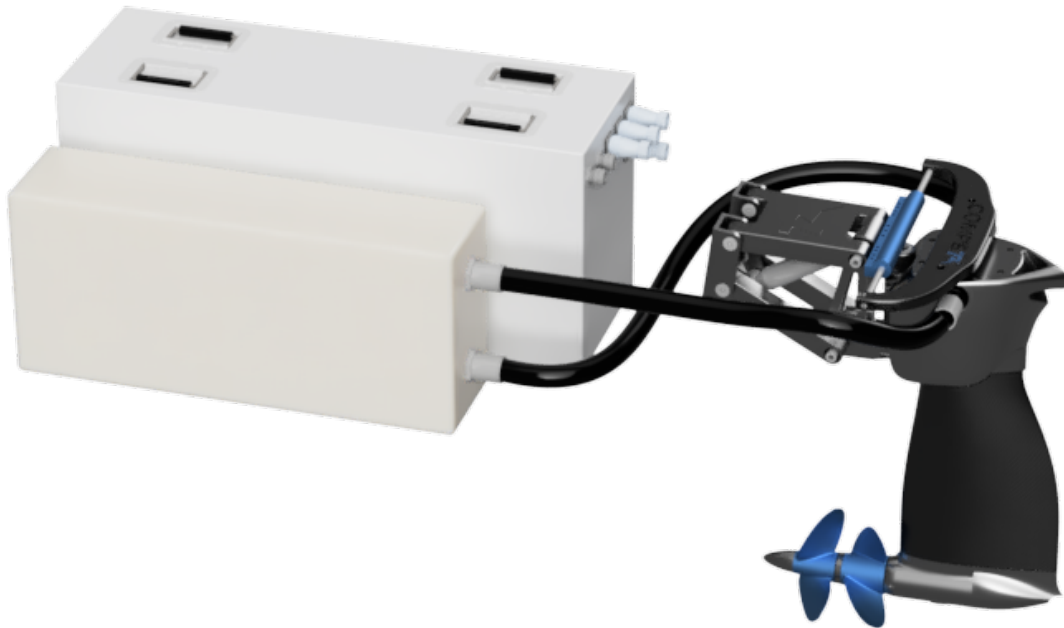
3. System Block Diagram



4. Powertrain description

5.1. General Description

The COMPETR powertrain is a patented electric outboard propulsion system based on a tractive, counter-rotating propeller architecture. The propeller arrangement and shaft geometry are designed to optimize hydrodynamic efficiency, thrust distribution, and maneuverability.



Example rendering of a possible layout

The low-profile motor head minimizes stern obstruction, freeing space on the transom and stern pulpit area typically occupied by internal combustion engines. The electric motor and gearbox are engineered for low acoustic emission, enabling smooth and quiet operation in applications requiring reduced noise levels.

5.2. Included Components

The COMPETR powertrain is delivered as an integrated system including the following components:

Mechanical and Propulsion System

- Counter-rotating propeller assembly
- Drive shaft with optimized hydrodynamic profile
- Gearbox and mechanical transmission components
- Outboard structural housing and mounting interfaces
- Hydraulic electric trim and hydraulic manual steering

Electrical Powertrain

- Electric motor

- Power inverter
- Battery pack
- High-voltage and low-voltage wiring harnesses

Thermal Management

- Integrated thermal management system for motor, inverter, and battery
- Cooling circuit and pumps

Control and Electronics

- Central controller for system supervision and power management

5.3. Optional Features and Configurations

The following features are available as optional equipment or configurable variants, subject to customer requirements and minimum order quantities (MOQ where applicable):

Steering

- Hydraulic electric steering system
- Steering position sensor

Control and Automation

- CAN bus interface for trim and steering pump control
- Autonomous-ready configuration when combined with steering position feedback

Battery and Energy Options

- Custom battery pack sizing and configuration
- On-board AC charging system (IEC 62196 Type 2 compliant)
- On-board 220 V inverter
- On-board solar charging capability

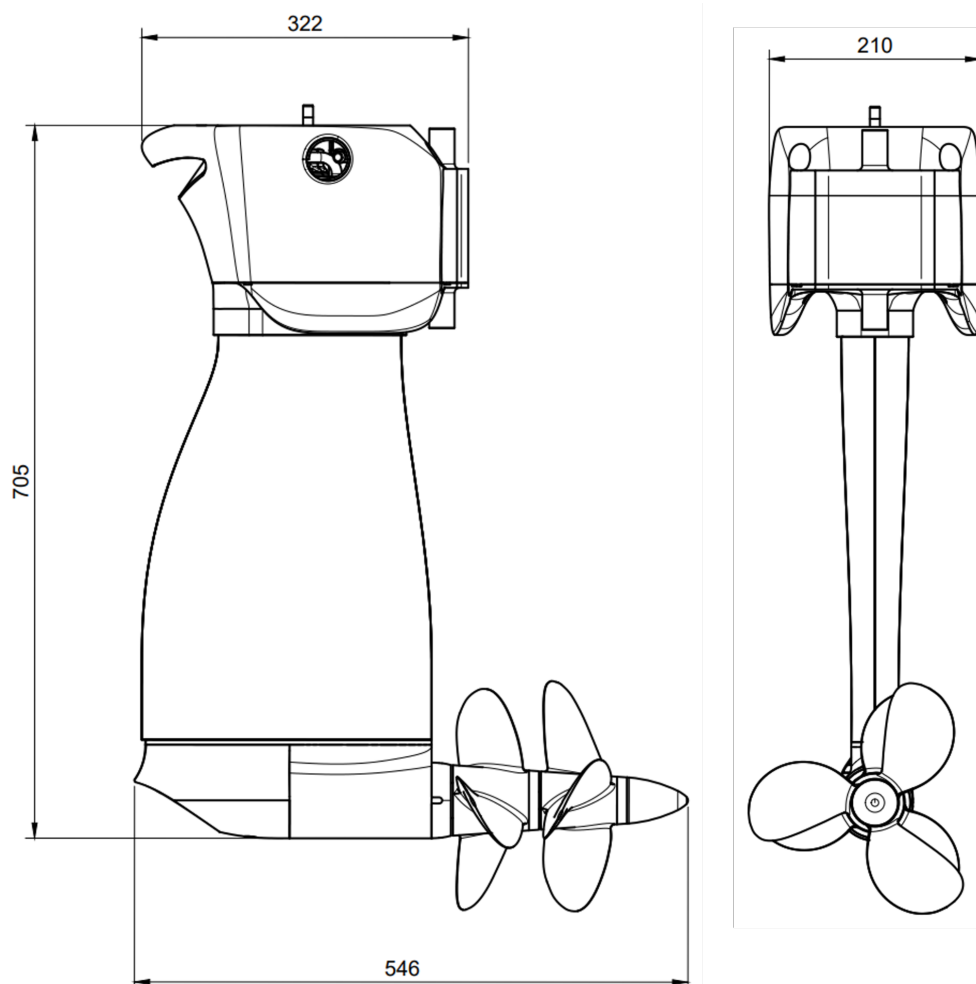
Connectivity and Telemetry

- Expansion modules enabling:
 - GSM
 - GPS
 - Bluetooth (BLE)
 - Wi-Fi
- Remote control, telemetry, diagnostics, and fleet management functionality

5. Recommended parts

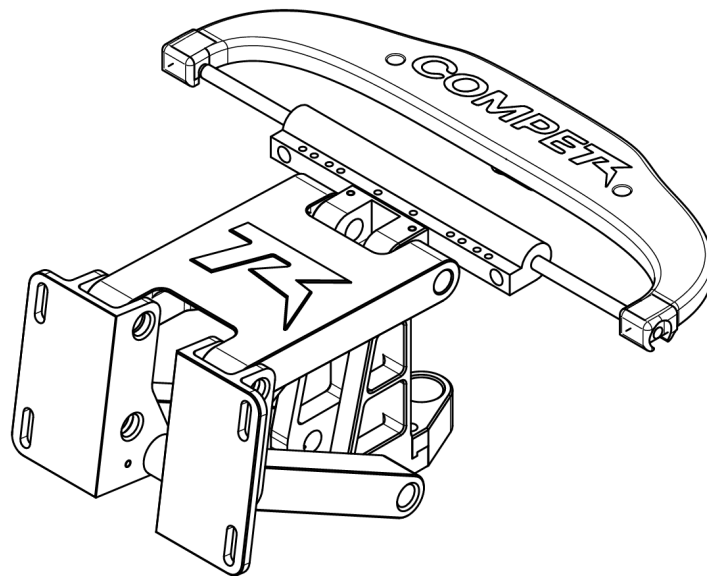
Item	P/N	Manufacturer	Notes
Inverter	On request	On request	Higher current rating required to reach full output power
Trim Pump	Hydraulic gear pump	On request	Refer to datasheet for best configuration to your hull. (flow, tank size, mounting...)
Steering Pump	Hydraulic gear pump	On request	
Cooling Pump	Marine pump	On request	
Bilge Pump	Marine pump	On request	

6. Outboard dimensions

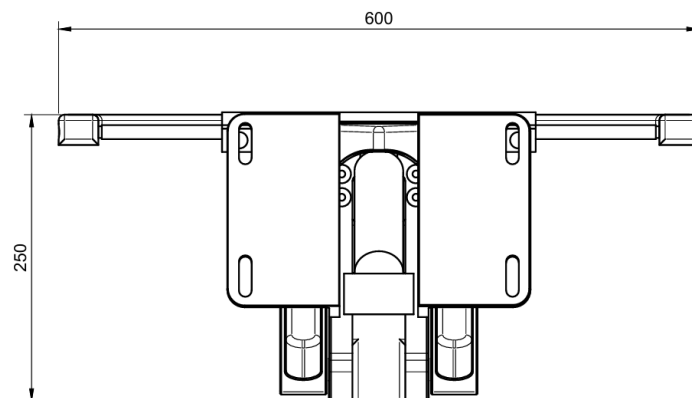


7. Bracket dimensions

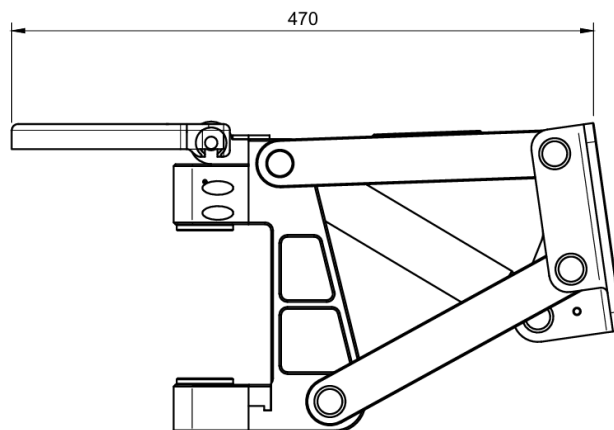
7.1. General view



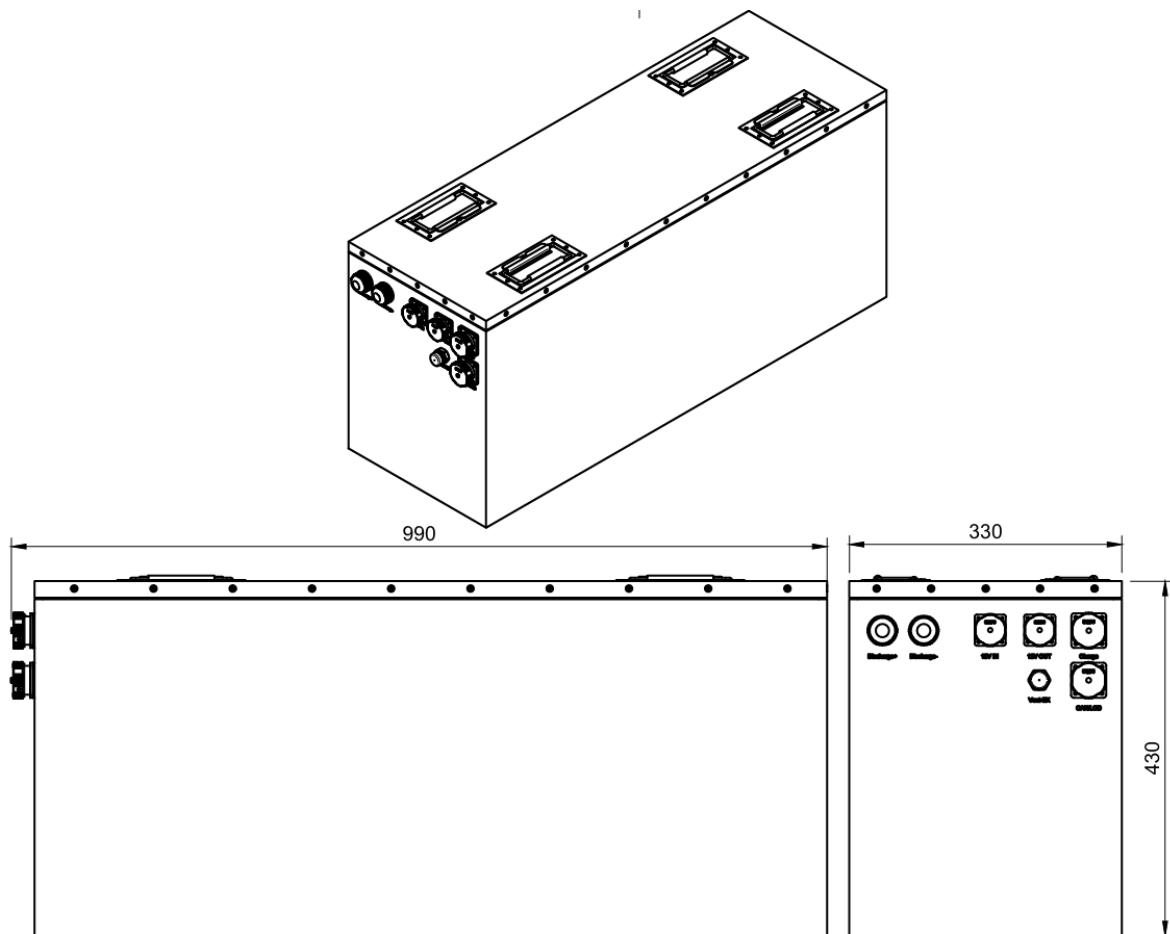
7.2. Rear view



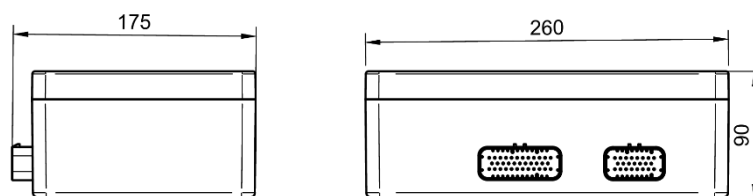
7.3. Side view



8. Battery pack dimensions

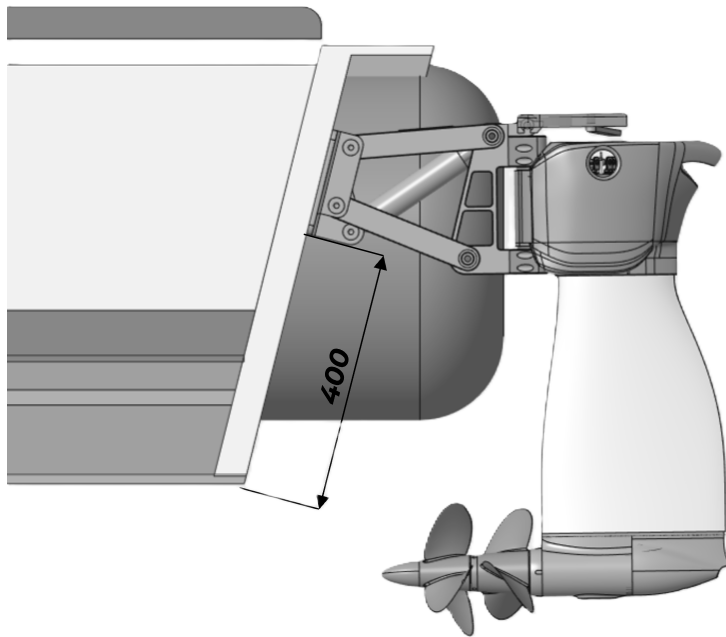


9. Control Unit dimensions



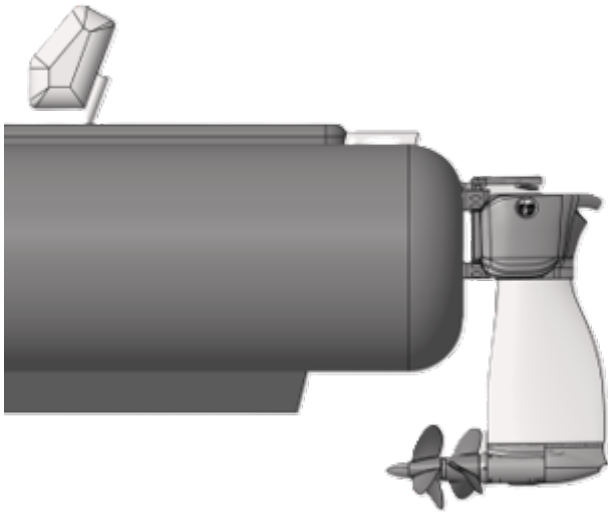
10. Transom

10.1. Mounting height (transom angle 14°)

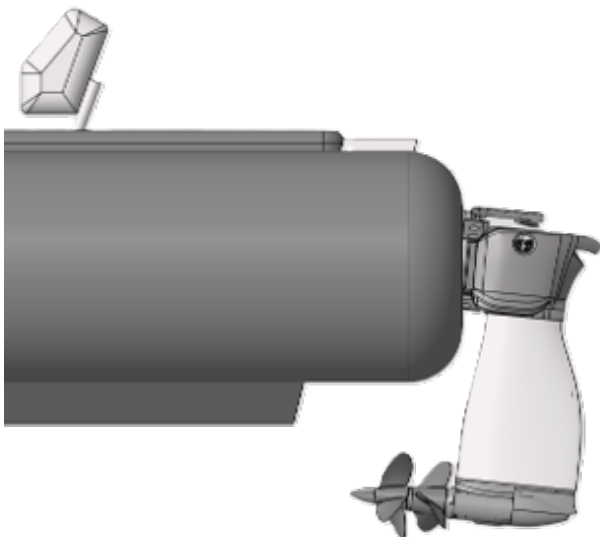


11. Trim angle

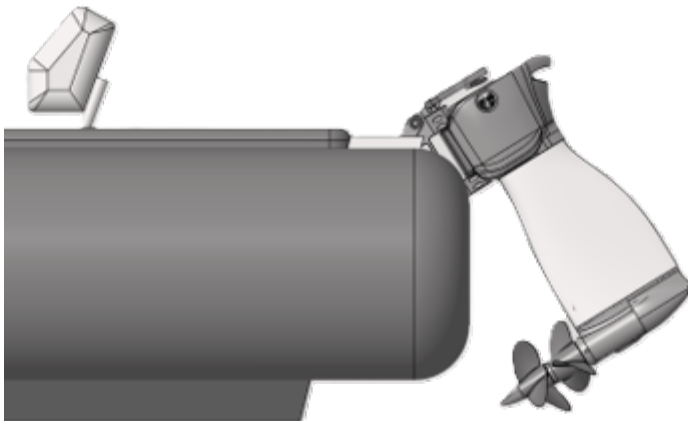
11.1. Neutral



11.2. Minimum [-5°]



11.3. Maximum [+30°]



12. Electrical connection

12.1. Encoder and Temperature sensor

Encoder side connector p/n: on request (mating p/n: on request)

Temperature sensor side connector p/n: on request (mating p/n: on request)

Male pin p/n: on request

Female pin p/n: on request

Recommended cable: shielded twisted-pair (equivalent grade)

Pin	Wire Color (Inverter side)	Signal Name	Notes	Wire Color (Encoder)	Inverter Pin
1	Brown	5V	Encoder Supply, TP with Pin#5	Red	3
2	Green	CLK+	SSI Clock Pos, TP with Pin#6	Black	11
3	Blue	DATA+	SSI Data Pos, TP with Pin#7	Grey/Pink	9
4	NC				
5	Brown/White	GND	Encoder GND, TP with Pin#1	Blue	2
6	Green/White	CLK-	SSI Clock Neg, TP with Pin#2	Violet	12
7	Blue/White	DATA-	SSI Clock Neg, TP with Pin#3	Red/Blue	10
8	NC				

Temperature sensor is an analog PTC type.

Pin	Wire Color	Signal Name	Notes	Inverter Pin
1	Orange	PTC+	Twisted pair	7
2	Orange/White	PTC-		1